

Echanny Emmanuel Idagu

Software Developer · Automation Engineer · Generative AI Engineer

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PROFESSIONAL SUMMARY

Full-stack developer and automation engineer with hands-on experience shipping production-grade web applications, RESTful API backends, and multi-node n8n automation pipelines integrated with Generative AI models including Google Gemini and OpenAI. Builds systems end-to-end — from MongoDB schema design and Express.js API architecture through to React frontend deployment on Vercel and Node.js backend deployment on Render. Delivered platforms handling authentication, RBAC, real-time scoring, OCR data extraction, and AI-driven analysis. Writes modular, maintainable ES6+ JavaScript guided by single-responsibility principles. Actively developing expertise in LLM prompt engineering, Retrieval-Augmented Generation (RAG) pipelines, and agentic AI workflow design.

TECHNICAL SKILLS

Languages & Frameworks:	JavaScript (ES6+), React.js (Hooks, Context API), Node.js, Express.js, HTML5, CSS3, Tailwind CSS
Backend & APIs:	REST API Design & Development, JWT Authentication, bcrypt, Middleware Architecture, Rate Limiting, CORS Configuration, Role-Based Access Control (RBAC)
Databases:	MongoDB (Atlas, Mongoose ODM), MySQL (Basic), Pinecone (Vector Database)
Automation & AI:	n8n Workflow Automation, Prompt Engineering, Google Gemini API, OpenAI API, OCR Integration, RAG Pipeline Design, AI Agent & Agentic Workflow Systems, Generative AI Integrations
Tools & Platforms:	Git & GitHub, Postman, Vercel, Render, Google Sheets API, Slack API
Soft Skills:	Systems Thinking, Problem Decomposition, Cross-functional Collaboration, Technical Documentation, Attention to Detail, Adaptability, Time Management

WORK EXPERIENCE

Software Developer Intern

Codveda Technologies · Internship · May 2026 – Present

- Developing and maintaining full-stack web application features using React.js and Node.js/Express.js, contributing directly to production codebase under senior engineering supervision.
- Building and integrating RESTful APIs with structured request validation, error handling middleware, and MongoDB data persistence using Mongoose ODM.
- Collaborating within an agile engineering team — participating in sprint planning, code reviews, and daily standups — gaining practical exposure to professional software delivery workflows.
- Applying automation and Generative AI integration skills to internal tooling tasks, including workflow design and AI-assisted feature prototyping.
- Documenting technical implementations and maintaining code quality standards consistent with team conventions, version control practices (Git branching, pull requests), and deployment procedures.

Volunteer Software Developer / Automation Engineer

Sannex Tech · Volunteer · Remote

- Contributed to backend API development including endpoint implementation, request validation, and third-party service integration, gaining direct exposure to production-level code standards and team review processes.
- Participated in structured debugging sessions — tracing API failure points, diagnosing logic errors, and documenting fixes — contributing to improved application stability and faster resolution of recurring issues.
- Collaborated with senior developers on development workflow practices including Git version control (branching, pull requests) and task coordination within an agile-adjacent team structure.
- Gained applied experience in backend architecture patterns, server-side request handling, and database query design within a structured, deadline-driven engineering environment.

PROJECT EXPERIENCE

ScoreFlow AI — AI-Augmented Submission & Scoring Platform

Software Developer & System Architect

- Designed and deployed a production full-stack platform centralising submission management and structured evaluation for hackathons, recruitment programmes, startup competitions, and innovation challenges.
- Engineered a modular Express.js backend with dedicated route layers for authentication, campaign management, submission handling, and real-time scoring, following single-responsibility principles across the codebase.
- Implemented JWT-based authentication with role-based access control (RBAC) supporting three distinct user roles — administrator, evaluator, and participant — each with clearly scoped permissions and protected route access.
- Configured MongoDB Atlas as the primary data store with Mongoose ODM schemas; resolved Render deployment issues including CORS, proxy trust configuration, and express-rate-limit compatibility.
- Built a responsive React 18 + Vite frontend with Tailwind CSS deployed to Vercel; integrated n8n automation for scheduled background evaluation workflows via long-expiry Bearer token authentication.

Tech Stack: React.js · Node.js · Express.js · MongoDB · JWT Authentication · Tailwind CSS · Vercel · Render

Career Coach Navigator — AI Resume Match & Gap Analysis Tool

Automation & Generative AI Engineer

- Built an AI-powered career assistant that accepts resume and job description inputs, then runs a multi-step n8n prompt chaining pipeline to analyse skill gaps, keyword alignment, and overall job-fit scoring.
- Designed layered Gemini API prompt chains with role-specific system instructions to generate structured gap analysis reports, ranked improvement recommendations, and tailored resume rewrite suggestions.
- Developed interactive React dashboards surfacing AI-generated match scores, missing keyword lists, and prioritised action items — formatted for both candidates and recruiting teams.
- Created reusable n8n workflow templates for career analysis pipelines, enabling rapid reconfiguration for different job description targets and candidate batches.

Tech Stack: React.js · Tailwind CSS · n8n · Google Gemini API · Prompt Chaining · Workflow Automation

Viral Loop Social Agent — AI Content Automation System

Automation Engineer

- Engineered an end-to-end n8n automation pipeline that accepts a content brief and autonomously generates platform-optimised social media captions, relevant hashtag sets, and visual content via the Stability AI image generation API.
- Integrated a Slack-based human-in-the-loop approval gate that routes generated content for team review before publishing, reducing unreviewed content from reaching production channels.
- Connected Google Sheets as a live content calendar and audit log, enabling teams to track published items, approval status, and content metadata without additional tooling overhead.
- Reduced manual content production touchpoints from five steps to one approval action through fully automated generation, routing, and logging.

Tech Stack: n8n · Google Gemini AI · Stability AI API · Slack API · Google Sheets

Receipt OCR & Expense Tracking Application

Backend Developer

- Built a receipt processing backend that accepts uploaded receipt images, passes them through an OCR API, and extracts structured transaction data (vendor, amount, date, category) for automatic storage and retrieval.
- Developed transaction categorisation logic and search/filter API endpoints, enabling users to query spending history by date range, merchant type, and expense category.
- Implemented JWT-secured authentication with per-user data isolation, protected transaction management routes, and input validation middleware.
- Designed a React analytics dashboard with spending summaries, filterable transaction history tables, and category-breakdown views for actionable financial insight.

Tech Stack: React.js · Node.js · Express.js · MongoDB · OCR APIs

EDUCATION & TRAINING

University of Calabar

BSc, Environmental Protection and Resources Management

Developed analytical, research, and systems-thinking skills applied to complex real-world problems — a foundation that directly informs how I approach software architecture, data modelling, and problem decomposition in engineering contexts.

Tech4Dev — MERN Stack Development

Full Stack Development Training

Completed project-based training in full-stack web development using MongoDB, Express.js, React.js, and Node.js (MERN). Built and deployed functional web applications with authentication, database integration, and RESTful API design.

TechCrush — Backend Development Programme

Backend Development Trainee

Hands-on backend training covering API architecture, server-side business logic, database design, middleware patterns, and application security using Express.js and MongoDB.

DeepTech Ready — Generative AI

Generative AI Fellow

Developed AI-driven automation projects and practical workflow systems, with focus on large language model integrations, AI agent design, and applying LLM APIs to real-world use cases.

Darey.io (3MTT) — Backend Development

Backend Development Training

Built backend systems and RESTful APIs using Node.js, Express.js, and both relational (MySQL) and non-relational (MongoDB) databases. Covered deployment workflows, environment configuration, and backend security fundamentals.

CERTIFICATIONS

- Enyata × Interswitch Buildathon Hackathon Certification — Recognised for project delivery and technical execution in a competitive, time-boxed innovation challenge.
- Generative AI Fellowship Certification — Awarded on practical delivery of LLM integration, RAG systems, and agentic workflow automation projects.
- MERN Stack Development Certification — Full-stack credential covering MongoDB, Express.js, React.js, and Node.js.
- Backend Development Certification — Server-side engineering covering APIs, authentication, database design, and deployment.
- Workflow Automation & AI Integration Training Certificate — n8n pipeline design, Generative AI API integration, and automated workflow architecture.

LEADERSHIP & COMMUNITY

- Competed in the Enyata × Interswitch Buildathon — a multi-team hackathon requiring rapid ideation, technical execution, and live presentation of a working solution under time pressure, building skills in project scoping, team coordination, and high-stakes delivery.
- Active contributor in developer and automation communities, sharing practical knowledge on n8n workflow design, Generative AI API integration, and full-stack deployment patterns.
- Advocates for building technology that is accessible and practically useful — approaching every project from the perspective of the end user's actual problem, not just the technical implementation.

CAREER OBJECTIVE

Seeking an internship or entry-level position as a Software Developer, Automation Engineer, or Generative AI Engineer where I can apply full-stack and AI-integration skills to real product challenges. I am looking for an environment that values engineers who take ownership, think in systems, and write code that solves actual problems — not just code that compiles. I bring hands-on experience with production deployments, multi-node automation pipelines, and LLM-integrated application development, and I am committed to deepening this expertise within a collaborative, technically rigorous team.

References available upon request.